

Project Preprocessing Phase

Date	1st july 2026
Team ID	SWTID-2026-6335
Project Name	HEART DISEASE ANALYSIS
Maximum Marks	5 Marks

Preprocessing Steps:

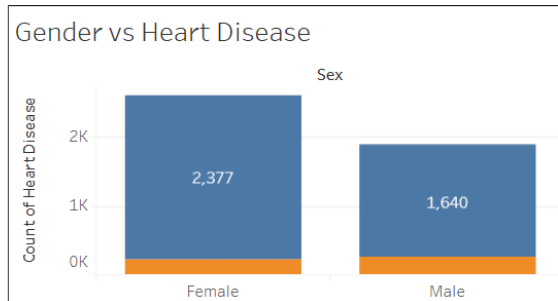
S.No	Step	Description
1.	Data Collection	Collected the Heart Disease dataset from a reliable public source (CSV format).
2.	Data Import	Imported the dataset into Tableau Desktop for analysis.
3.	Data Inspection	Examined all columns, data types, and identified missing or inconsistent values.
4.	Handling Missing Values	Verified the dataset for null values and removed or replaced missing records where necessary.
5.	Removing Duplicate Records	Checked for duplicate entries and eliminated duplicates to maintain data integrity.

6.	Data Type Validation	Ensured numerical fields (Age, Cholesterol, Blood Pressure, Maximum Heart Rate) and categorical fields (Gender, Chest Pain Type, Heart Disease) were correctly formatted.
7.	Data Cleaning	Standardized categorical values (e.g., Male/Female, Yes/No) and corrected inconsistencies.
8.	Feature Engineering	Created calculated fields such as Heart Disease Percentage, Average Cholesterol, and Patient Count.
9.	Data Validation	Verified cleaned data for accuracy before visualization.
10.	Data Visualization Preparation	Connected processed data to Tableau worksheets and created dimensions, measures, filters, and hierarchies.

Business Questions with Visualizations:

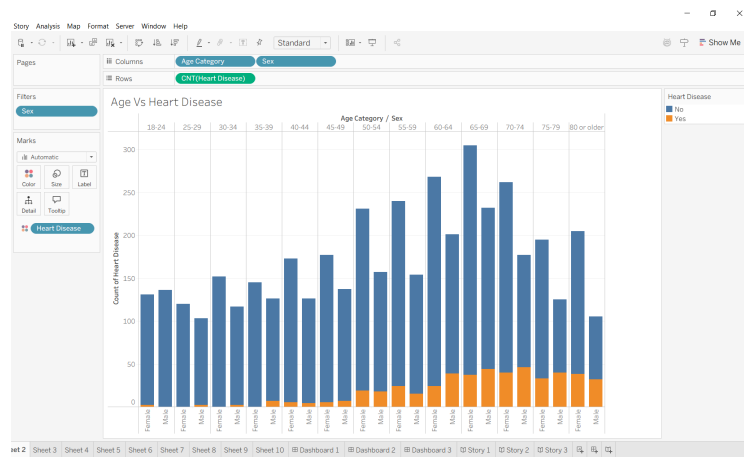
1.How is heart disease distributed across genders?

A.Female participants make up a slightly larger share of the dataset. Heart disease is present in both males and females, showing that gender alone is not a sufficient predictor.



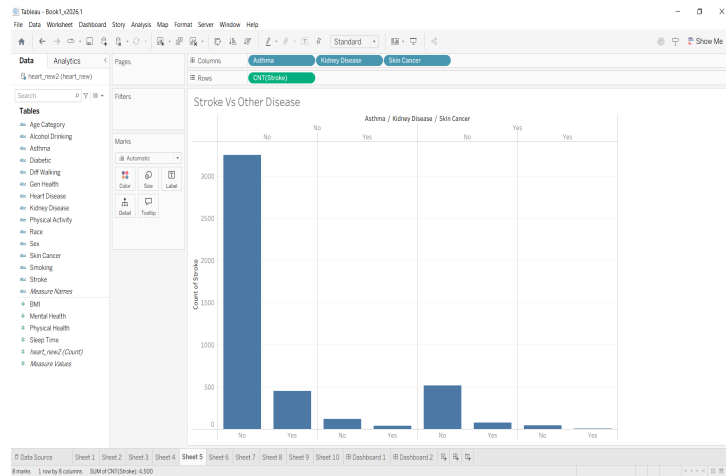
2. Which age groups are most affected by heart disease?

A. Heart disease cases increase steadily with age. Individuals aged **60 years and above**, particularly those **80 or older**, have the highest number of cases.



3. Do other chronic diseases increase stroke risk?

A. Stroke cases are relatively higher among patients with kidney disease and other chronic illnesses. Multiple chronic conditions increase overall cardiovascular risk.



4. What are the key performance indicators (KPIs) of the dashboard?

A. The dashboard summarizes the total heart disease cases, gender distribution, race-wise distribution, smoking and alcohol impact, and general health status, providing an overview of cardiovascular risk factors.

5. What is the overall conclusion from the dashboard? Heart disease is influenced by multiple interconnected factors. Age is the strongest predictor, while poor general health, smoking, diabetes, higher BMI, and chronic diseases all contribute to an increased risk of heart disease and stroke.

